

Worm Gear Reducers Installation, Lubrication and Maintenance Instructions

Selection Information

Read ALL instructions prior to operating reducer. Injury to personnel or reducer failure may be caused by improper installation, maintenance or operation.

Written authorization from GROVE GEAR / ELECTRA-GEAR is required to operate or use reducers in man lift or people moving devices.

Check to make certain application does not exceed the allowable load capacities published in the current catalog.

Buyer shall be solely responsible for determining the adequacy of the product for any and all uses to which Buyer shall apply the product. The application by Buyer shall not be subject to any implied warranty of fitness for a particular purpose.

Safety Alert

WARNING

- For safety, Buyer or User should provide protective guards over all shaft extensions and any moving apparatus mounted thereon. The User is responsible for checking all applicable safety codes in his area and providing suitable guards. Failure to do so may result in bodily injury and/or damage to equipment.
- Hot oil and reducers can cause severe burns. Use extreme care when removing lubrication plugs and vents.
- Make certain that the power supply is disconnected before attempting to service or remove any components. Lock out the power supply and tag it to prevent unexpected application of power.
- Reducers are not to be considered fail safe or self-locking devices. If these features are required, a properly sized, independent holding device should be utilized. **Reducers should not be used as a brake.**
- Any brakes that are used in conjunction with a reducer must be sized or positioned in such a way so as to not subject the reducer to loads beyond the catalog rating.
- Lifting supports including eyebolts are to be used for vertically lifting the gearbox only, with no other associated attachments or motors.
- Use of an oil with an EP additive on units with backstops may prevent proper operation of the backstop. Injury to personnel, damage to the reducer or other equipment may result.
- Overhung loads subject shaft bearings and shafts to stress which may cause premature bearing failure and/or shaft breakage from bending fatigue, if not sized properly.

CAUTION

- Test run unit to verify operation. If the unit tested is a prototype, that unit must be of current production.
- If the speed reducer cannot be located in a clear and dry area with access to adequate cooling air supply, then precautions must be taken to avoid the ingestion of contaminants such as water and the reduction in cooling ability due to exterior contaminants.
- Mounting bolts should be routinely checked to ensure that the unit is firmly anchored for proper operation.

Important Information

In the event of the resale of any of the goods, in whatever form, Resellers/Buyers will include the following language in a conspicuous place and in a conspicuous manner in a written agreement covering such sale:

The manufacturer makes no warranties or representations, express or implied, by operation of law or otherwise, as to the merchantability or fitness for a particular purpose of the goods sold hereunder. Buyer acknowledges that it alone has determined that the goods purchased hereunder will suitably meet the requirements of their intended use. In no event will the manufacturer be liable for consequential, incidental or other damages. Even if the repair or replacement remedy shall be deemed to have failed of its essential purpose under Section 2-719 of the Uniform Commercial Code, the manufacturer shall have no liability to Buyer for consequential damages.

Resellers/Buyers agree to also include this entire document including the warnings above in a conspicuous place and in a conspicuous manner in writing to instruct users on the safe usage of the product.

This instructions manual should be read together with all other printed information such as catalogs, supplied by Grove Gear / Electra-Gear.

General Operation

1. Run the motor which drives the reducer and check the direction of reducer output rotation. Consult motor nameplate for instructions to reverse the direction of rotation.
2. Attaching the load: On direct coupled installations, check shaft and coupling alignment between speed reducer and loading mechanism. On chain/sprocket and belt/pulley installation, locate the sprocket or pulley as close to the oil seal as possible to minimize overhung load. Check to verify that the overhung load does not exceed specifications published in the catalog.
3. High momentum loads: If coasting to a stop is undesirable, a braking mechanism should be provided to the speed reducer output shaft or the driven mechanism.

CAUTION

The system of connected rotating parts must be free from critical speed, torsional or other type vibration, no matter how induced. The responsibility for this system analysis lies with the purchaser of the speed reducer.

Installation

1. Mount the unit to a rigid flat surface using grade 5 or higher fasteners. The mounting fasteners should be the largest standard size that will fit in the base mounting hole. Shim as required under flange or base feet which do not lie flat against the mounting surface.
2. For shipment, pipe plugs are installed in the unit and a vent plug is packed separately. After mounting the unit in position, remove the appropriate pipe plug and install the vent plug in the location shown on page 5. On double reduction units both the primary and the secondary must be vented. Failure to vent the unit can cause premature seal wear or loss of seal and oil. These conditions are not covered by warranty. Check for correct oil level. Contact the Factory for level and vent recommendations on non-standard mounting positions.
3. Connect motor to speed reducer.

Depending upon gear geometry and operating conditions worm gear reducers may or may not back-drive. Use of a brake or external holding device is required if any evidence of backdriving is not desired.

Special consideration should be given to high inertia loads connected to the output shaft. Consult the factory for further details.

⚠ WARNING DO NOT CHANGE MOUNTING POSITIONS WITHOUT CONTACTING FACTORY. Altering the mounting position may require special lubrication provisions which must be factory installed.

⚠ CAUTION Do not operate the reducer without making sure it contains the correct amount of oil. Do not overfill or underfill with oil, or injury to personnel, reducer or other equipment may result.

⚠ CAUTION A unit cannot be used as an integral part of a machine superstructure which would impose additional loads on the unit other than those imposed by the torque being transmitted either through a shaft-mounted arrangement, and any shaft mounted power transmitting device. (*e.g., sprockets, pulleys, couplings*)

⚠ CAUTION For safe operation and to maintain the unit warranty, when changing a factory installed fastener for any reason, it becomes the responsibility of the person making the change to properly account for fastener grade, thread engagement, load, tightening torque and the means of torque retention.

Lubrication - Standard and WASHGUARD® / Platinum Units

All standard reducers ordered from Factory are filled with Mobil Glygoyle 460 polyglycol (**PAG**) lubricant or equivalent suitable for continuous operation within a -10° F to 120° F ambient temperature range. Double and triple reduction units have separate oil sumps and must be filled/checked independently. Prior to startup, verify that the oil is at the level shown on page 6. Lubricant type is stamped on all nameplates.

Change Intervals: Standard compounded lubricants (non-synthetic) should be changed every six months or 2500 operating hours, whichever comes first. Factory installed synthetic lubricants should be changed only when performing maintenance that requires gearbox disassembly.

CAUTION Oil should be changed more often if reducer is used in a severe environment (i.e. dusty, humid).

CAUTION In the Food and Drug Industry (including animal food), consult the lubrication supplier for recommendation of lubricants which are acceptable to the Food and Drug Administration and/or other authoritative bodies having jurisdiction. Factory supplied **PAG** oil is acceptable for incidental food contact (NSF H1) for use in and around food processing areas.

CAUTION Do not mix different oils in the reducer. Grove Gear / Electra-Gear reducers are shipped standard with **PAG** lubricant – this lubricant is not compatible with conventional mineral or PAO synthetic oils.

Special Lubrication Requirements - Size 818 and Larger

Units shipped from Factory are assembled to properly lubricate all internal components based on a specific assumed mounting orientation. If a size 818 or larger unit will be mounted in an orientation different from its original intension, as shown in product catalog, or run with sustained input speeds less than 1200 RPM, it should be specified with the order. The unit can then be modified to assure proper lubrication.

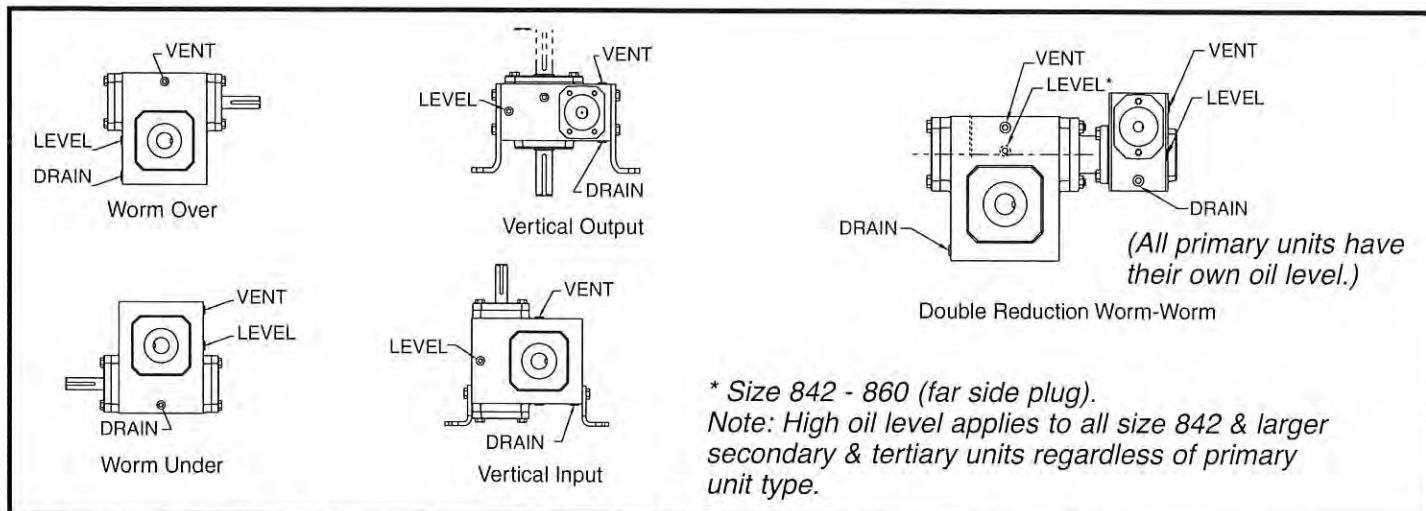
The precision-made gears and bearings in Grove Gear / Electra-Gear Speed Reducers require high-grade lubricants of the proper viscosity to maintain trouble-free performance. All standard reducers ordered from the factory are filled with ISO viscosity grade 460 polyglycol (**PAG**) lubricant. If oil needs to be added or changed, ONLY compatible polyglycol lubricants should be used. Contact the factory for more information.

Oil Capacities (ounces) - Standard Units

Mounting Position	UNIT SIZE													
	813	815	818	821	824	826	830	832	842	852	860	870*	880*	8100*
Worm Over	4	12	12	20	24	40	56	72	112	188	312	560	768	1152
Worm Under	8	16	20	28	40	60	84	108	152	304	328	524	820	1280
Vertical Output	4	16	16	28	32	48	68	88	128	248	320	332	460	640
Vertical Input	4	16	16	24	32	48	72	92	128	248	325	584	800	1200
Extended Bearing	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8	12	17	27	640	1008	1632
Worm Over on Secondary Unit of Double Reduction	N/A	N/A	N/A	N/A	N/A	N/A	N/A	192	308	320	485	805	114	1716
Stainless Steel + WASHGUARD® - All Mounting Positions	6	14	18	26	32	50	78	98	N/A	N/A	N/A	N/A	N/A	N/A

CAUTION Always check for proper oil level after filling. Capacities vary somewhat with model and mounting position. Oil should rise to bottom edge of level hole. Do not overfill.

Standard Gear Reducer Mounting Positions & Vent Plug, Level and Drain Locations ▲



▲ Does not apply to Stainless Steel, WASHGUARD® or units with Enviroseal option.

Maintenance - Standard Units

Your Grove Gear / Electra-Gear reducer has been tested and adjusted at Factory. Dismantling or replacement of components must be done by Grove Gear / Electra-Gear to maintain the warranty.

1. Frequently check the oil level of the reducer. If oil level is low, (refer to reducer vent and level position chart) add proper lubrication through the filler plug until it comes out the oil level plug.
2. Inspect vent plug often to insure it is clean and operating.
3. Always check for proper oil level after filling. Do not overfill or underfill with oil, or injury to personnel, reducer, or other equipment may result.
4. Do not mix different oils in the reducer.

Seals: The Grove Gear / Electra-Gear line of speed reducers utilizes premium quality seals which are the state-of-the-art in sealing technology. Seals are, however, a wear item and eventually need to be replaced. Replacement can be easily accomplished by following the steps below:

1. Remove the worn seal without damaging the shaft surface or the seal bore. This can be done by drilling a .062 diameter hole in the seal casing (being careful not to drill into the bearing behind the seal). Install a #10 sheet metal screw into the hole and pry out the seal.
2. Clean the seal bore of sealant.
3. Before installing the new seal, use electrical tape to cover any keyways on the shaft to prevent seal lip damage.
4. Grease the seal lips with bearing grease and apply a sealant to the seal bore.
5. Slide the seal into the shaft being careful not to fold the inner lip over on any shaft steps.
6. Press the seal into its bore with a sleeve that presses on the seal casing, being careful to keep the seal square in its bore.

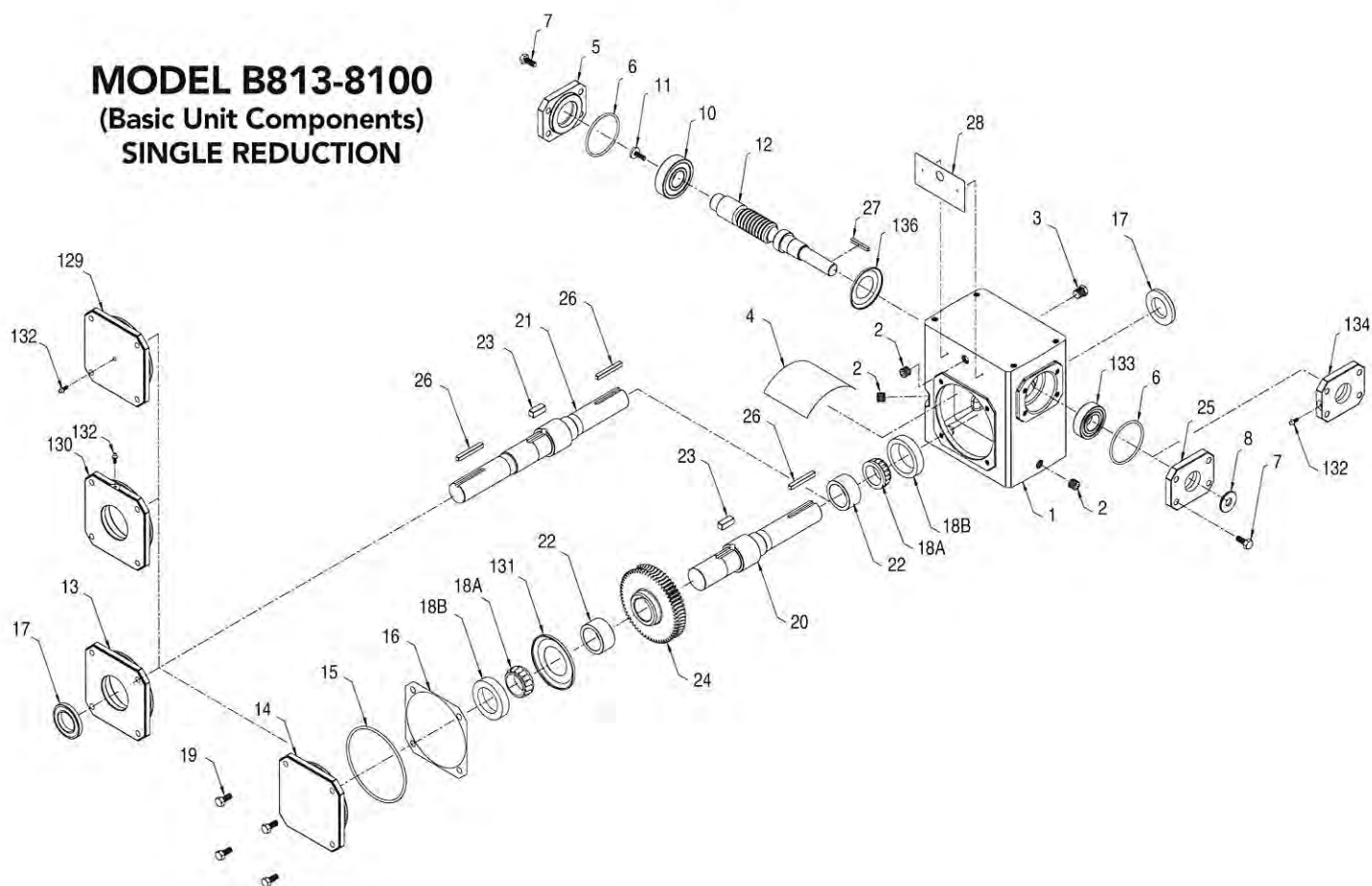
Class of Service

Load conditions must be within cataloged ratings published in the current Grove Gear / Electra-Gear Catalog (available upon request).

Published ratings assume lubrication with ISO 460 viscosity grade polyglycol (**PAG**) oil. Contact Factory for ratings when an alternate lubricant is used.

Warranty From Grove Gear / Electra-Gear - See 8050 catalog for warranty terms and conditions.

PARTS LIST

MODEL B813-8100
(Basic Unit Components)
SINGLE REDUCTION

BASIC SINGLE REDUCTION UNIT
(B-STYLE)

ITEM # DESCRIPTION

- 1 HOUSING
- 2 PIPE PLUG
- 3 VENT PLUG
- 4 SPLASH GUARD
- 5 INPUT CAP
- 6 O-RING
- 7 HEX HEAD CAP SCREW
- 8 INPUT OIL SEAL
- 9 INPUT BEARING (cup and cone for 842 and larger units)
- 10 INPUT BEARING (cup and cone for 842 and larger units)
- *11 RETAINING SCREW
- 12 INPUT WORM SHAFT
- 13 OUTPUT COVER - OPEN
- 14 OUTPUT COVER - CLOSED
- 15 O-RING

- 16 OUTPUT COVER SHIM (as required)
- 17 OUTPUT OIL SEAL
- 18 OUTPUT BEARING (18A. CONE, 18B. CUP)
- 19 HEX HEAD CAP SCREW
- ♣ 20 OUTPUT SHAFT - SINGLE
- ♣ 21 OUTPUT SHAFT - DOUBLE
- 22 GEAR SPACER
- 23 GEAR KEY (only used on size 826 and larger units)
- ♣ 24 OUTPUT GEAR
- *25 INPUT COVER
- 26 KEY - OUTPUT EXTENSION
- 27 KEY - INPUT EXTENSION
- 28 NAMEPLATE

(supplied only when mounting position involves a vertical shaft)

- *129 OUTPUT COVER - CLOSED
- *130 OUTPUT COVER - OPEN
- *131 OUTPUT BEARING GREASE RETAINER
- 132 GREASE FITTING
- 133 SEALED BALL BEARING (only used on size 818 thru 826 units)
- ♦134 INPUT COVER
- ♦136 INPUT BEARING GREASE RETAINER

* ONLY USED ON SIZE 842 AND LARGER UNITS

♦ ONLY USED ON SIZE 830 AND LARGER UNITS

♣ SUPPLIED ONLY AS OUTPUT ASSEMBLY ON 813 THROUGH 824 UNITS

VERTICAL SHAFT REQUIRED PARTS

Complete information is available online at www.grovegear.com

Continued on next page

GROVE GEAR

www.grovegear.com

PARTS LIST (cont'd)

QUILL MOTOR FLANGE UNIT
(BMQ-STYLE)

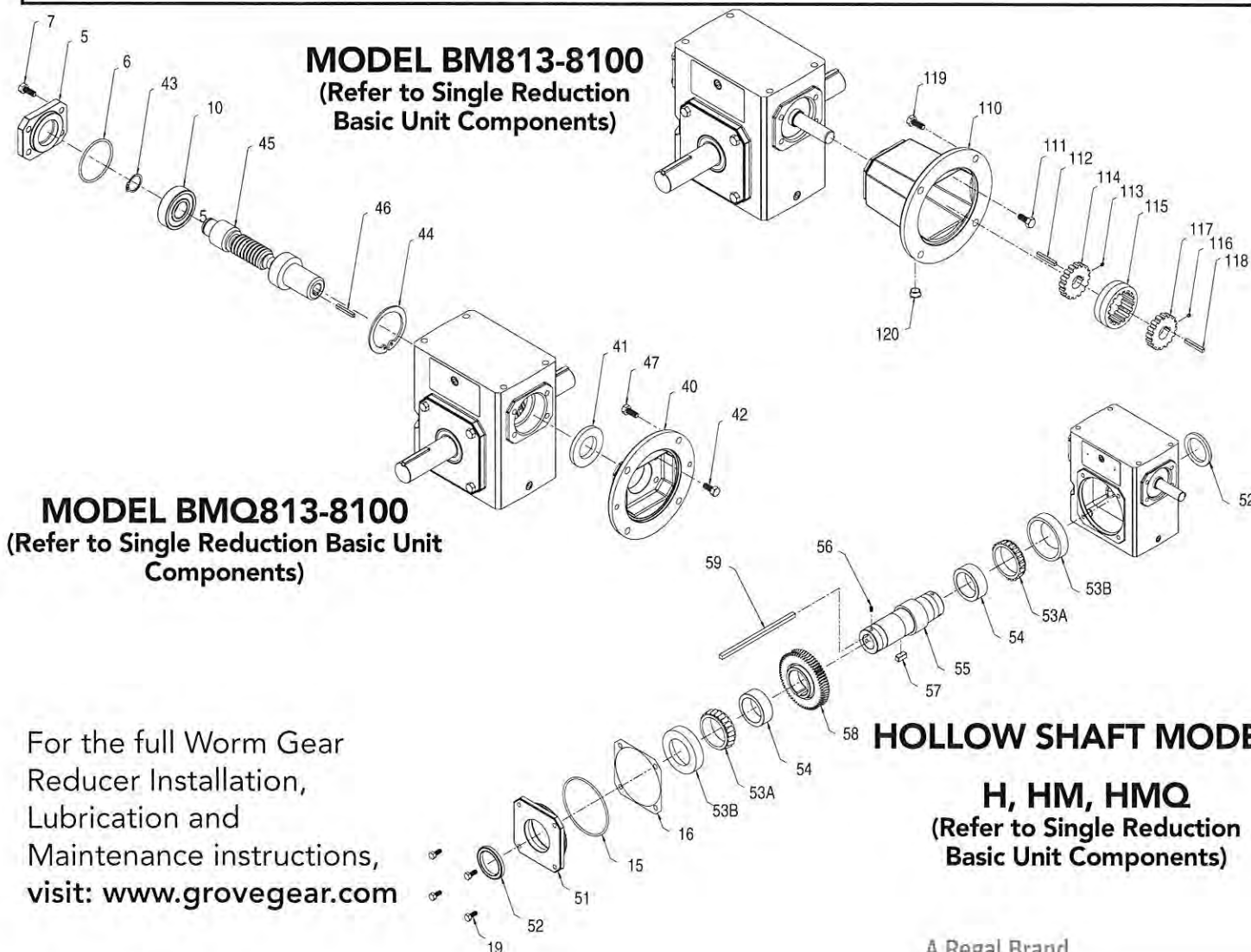
- 40 QUILL MOTOR FLANGE
- 41 INPUT OIL SEAL
- 42 HEX HEAD CAP SCREW (flange to housing)
- 43 RETAINING RING - SHAFT
- *44 RETAINING RING - HOUSING
- 45 QUILL INPUT SHAFT
- 46 KEY - INPUT
- 47 HEX HEAD CAP SCREW (motor to flange)

- 51 OUTPUT COVER
- 52 OUTPUT OIL SEAL
- 53 OUTPUT BEARING (53A. CONE, 53B. CUP)
- 54 GEAR SPACER
- ♣ 55 OUTPUT SHAFT
- 56 SETSCREW
- 57 GEAR KEY (only used on size 826 and larger units)
- ♣ 58 OUTPUT GEAR
- 59 OUTPUT KEY

- 111 HEX HEAD CAP SCREW (flange to housing)
- 112 COUPLING KEY - REDUCER SHAFT
- 113 SETSCREW - REDUCER SHAFT
- 114 COUPLING GEAR - REDUCER SHAFT
- 115 COUPLING SLEEVE
- 116 SETSCREW - MOTOR SHAFT
- 117 COUPLING GEAR - MOTOR SHAFT
- 118 COUPLING KEY - MOTOR SHAFT
- 119 HEX HEAD CAP SCREW (motor to flange)
- 120 PLASTIC PLUG

HOLLOW OUTPUT SHAFT UNIT
(H-STYLE)LONG MOTOR FLANGE AND
COUPLING KIT (BM-STYLE)

- 110 "C" FACE MOTOR FLANGE



HOLLOW SHAFT MODELS

H, HM, HMQ
(Refer to Single Reduction
Basic Unit Components)

A Regal Brand

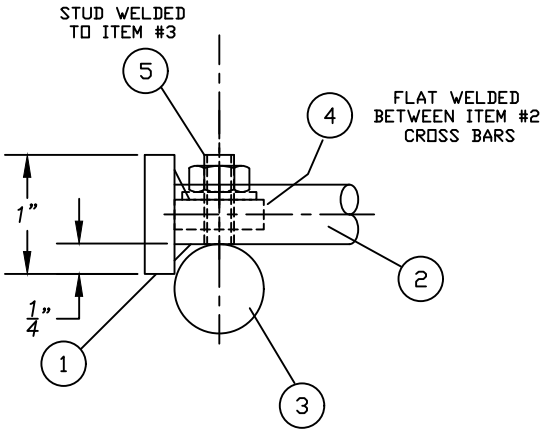
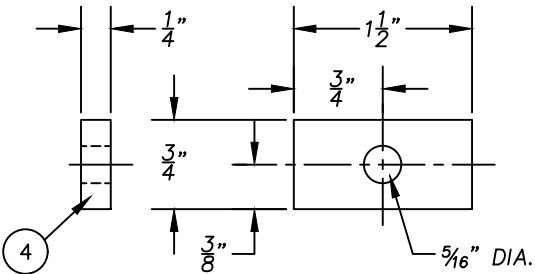
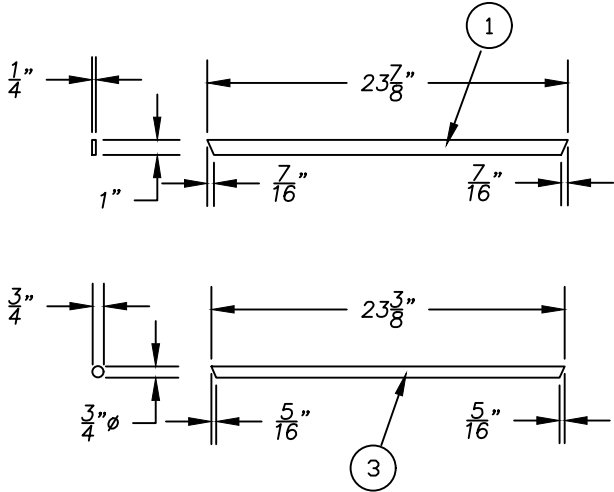
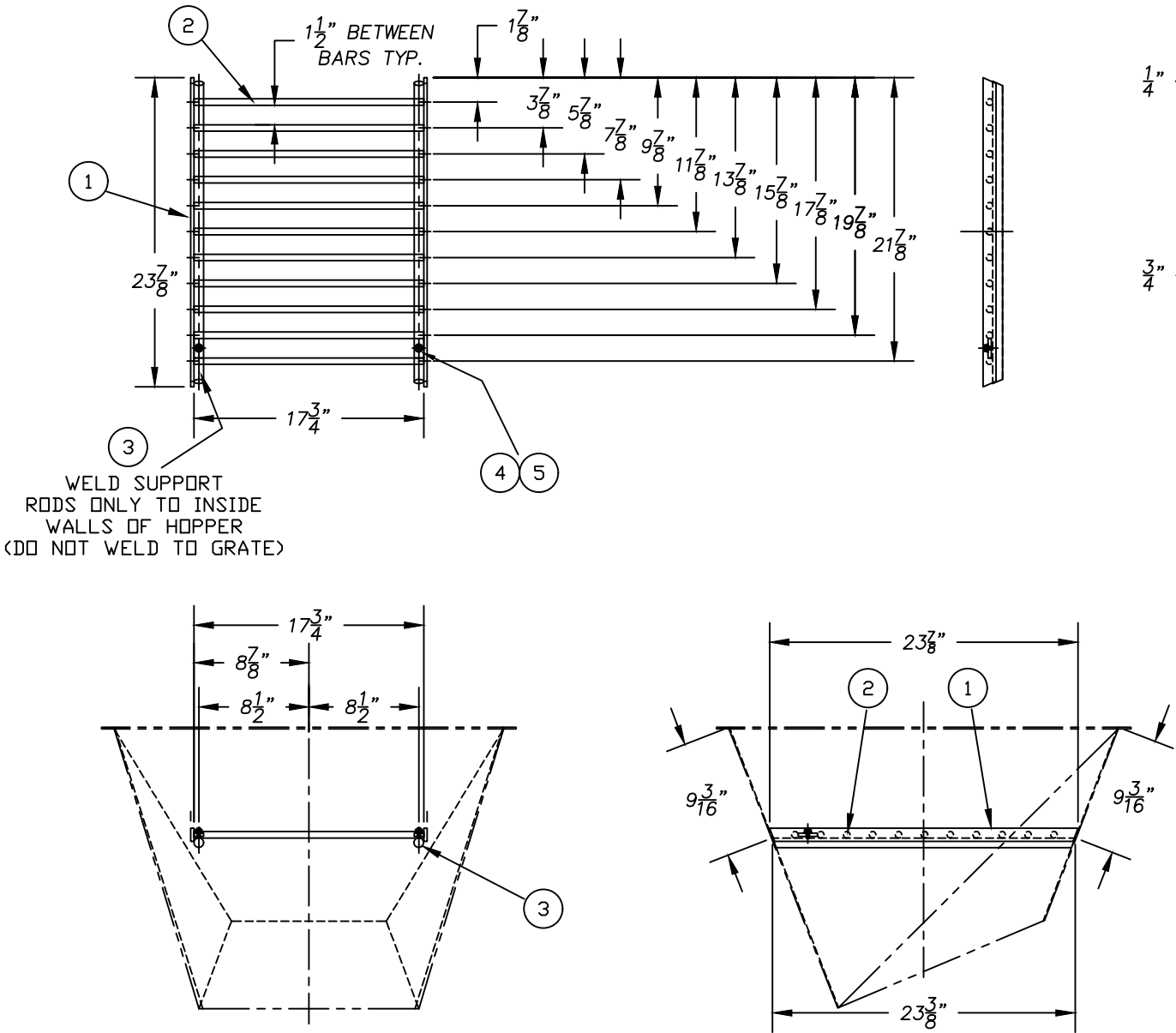
REGAL

www.regalbeloit.com

For the full Worm Gear
Reducer Installation,
Lubrication and
Maintenance instructions,
visit: www.grovegear.com

Grove Gear
Union Grove, Wisconsin 53182
PH: 262-878-1221
FAX: 262-878-1968

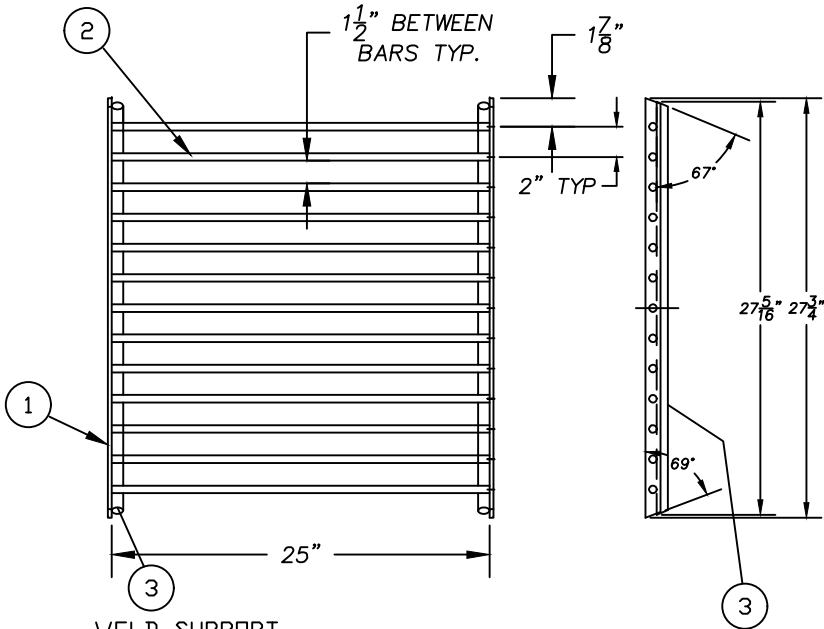
REV	DESCRIPTION	DATE	BY
A	ITEM NO. 4 WAS 1/2" THK.	8/26/2024	MP



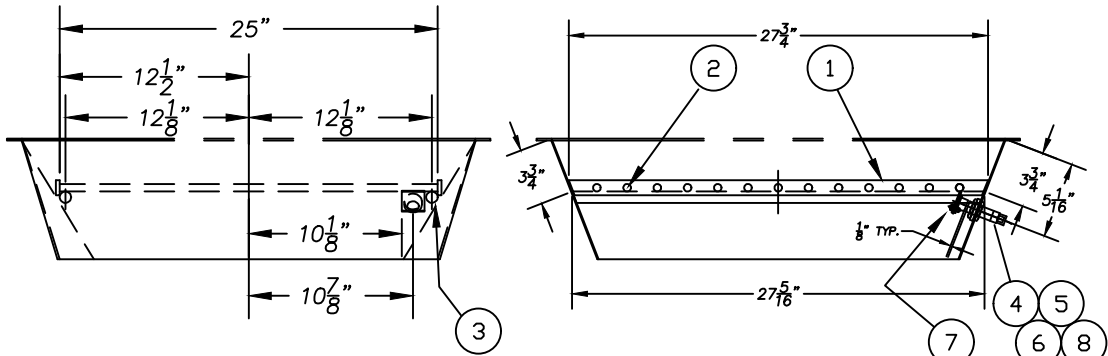
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE ENGLISH (FEET AND/OR INCHES) TOLERANCES			
DECIMALS	INCHES	FEET	ANGLES
.X ±.030		X'-X"=±1/4" UP TO 20 FEET	± 1°
.XX ±.015	X"= ±1/16"	X'-X"=±1/2" OVER 20 FEET	
.XXX ±.005			
DO NOT SCALE DWG. BREAK ALL SHARP EDGES			
MATERIAL			
FINISH			

HAPMAN Ideas that move			
P.O. BOX 2321 6002 E. KILGORE ROAD, KALAMAZOO, MICHIGAN 49048 PHONE (269) 343-1675 FAX (269) 349-2477			
DATE	8/26/16	JOB NO.	
SCALE	1/8" = 1"	BOM = 1	DWG = 1
TITLE	BAR GRATE ASSEMBLY BOLT IN PLACE, (LOW IN HOPPER), 304 SS		
CUST.			
SHEET	2 OF 2	DWG. NO.	X15851AA-C01
REV.	A		

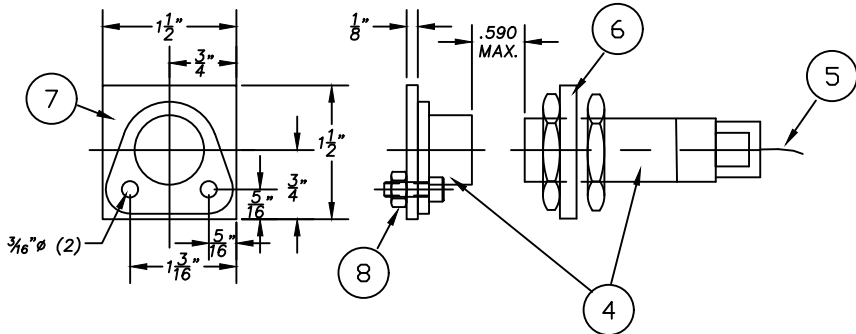
REV	DESCRIPTION	DATE	BY
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WELD SUPPORT
RODS ONLY TO INSIDE
WALLS OF HOPPER
(DO NOT WELD TO GRATE)



NOTE: ITEM #7 MUST
BE PARALLEL TO
HOPPER SIDEWALL



10			
9			
8		8-32 x 1/2 LG SHCS W/LOCKNUT, SS	2
7		TAB- 1/8 x 1 1/2 x 1 1/2, SS	1
6	38600	WASHER- 3/16 THK WHITE NEOP.	1
5	889D-F8AB-10	CABLE- 10 METER QC, AB	1
4	440N-Z21516H	SAFETY SENSOR & ACTUATOR- 18 MM, AB	1
3		SUPPORT BAR- 3/4ø x 27 1/2 LG, SS	2
2		BAR- 1/2ø x 25 LG, SS	13
1		END BAR- 1/4 x 1 x 28 LG, SS	2
ITEM NO.	DRAWING NO.	DESCRIPTION	QUAN

HAPMAN Ideas that move P.O. BOX 2321 6002 E. KILGORE ROAD, KALAMAZOO, MICHIGAN 49048 PHONE (269) 343-1675 FAX (269) 349-2477			
DATE	8/26/16	JOB NO.	
SCALE	1/8" = 1"	BOM = 0	DWG = 1
TITLE	BAR GRATE ASSEMBLY LIFT OUT, 30 SQ		
CUST.			
SHEET	1 OF 1	DWG. NO.	X15851AA-C02
		REV.	

UNLESS OTHERWISE SPECIFIED				
DIMENSIONS ARE ENGLISH (FEET AND/OR INCHES)				
TOLERANCES				
DECIMALS	INCHES	FEET	ANGLES	
.X ±.030		X'-X"=±1/4" UP TO 20 FEET		
.XX ±.015	X"= ±1/16"	X'-X"=±1/2" OVER 20 FEET	±1°	
.XXX ±.005				
DO NOT SCALE DWG.				
BREAK ALL SHARP EDGES				
MATERIAL				
FINISH				



Description

When it comes to machine safety, Rockwell Automation knows that protection of personnel and equipment is your main concern. At the same time, flexibility and productivity are points that must also be considered as you design your safety system. Optimize all of these with the new Allen-Bradley SensaGuard family of noncontact switches.

Featuring the latest generation of RFID technology for coding and inductive technology for sensing, SensaGuard's large sensing range and tolerance to misalignment is a cost-effective solution that is ideally suited for a wide range of industrial safety applications.

The SensaGuard product line is a Category 4 /SIL 3 rated switch per EN954-1, TÜV functional safety approved to IEC 61508.

Features

- Switches can be connect to a standard safety relay, for example, the MSR126, MSR127, MSR200/300 Family, SmartGuard and Safety I/O Blocks
- Multiple actuator sizes for large sensing distance
- IP 69K environmental rating
- Short-circuit and over-voltage protection
- Led located on the switch for door status and troubleshooting

Benefits

- No dedicated controller required
- Cat 4/SIL 3 rating maintained even with multiple units connected in series
- Switches can be connected in series with other devices (light curtain, E-stops, key interlock switches)
- Extended diagnostics for easy troubleshooting
- Large sensing distances
- Tolerance to misalignment
- Multiple sensing directions
- Stainless steel version suitable for use in harsh environments
- Use standard proximity brackets

Specifications

Category	Cat. 4/SIL3		
Certifications	TÜV, CE, cULus (UL 508)		
Standards	IEC 60947-5-3, IEC 61508, EN 954		
PFHd	1.119...10 ⁻⁹		
Operating Characteristics			
Sensing Distance (Target)	18 mm Plastic	30 mm Plastic	18 mm SS
Operating Distance, Make—mm (in)	Assured: 15 (0.59)	Assured: 25 (0.98)	Assured: 10 (0.39)
Case Material	Valox® DR 48		304 Stainless Steel
Actuator Material	Valox® DR 48		304 Stainless Steel
Typical Misalignment	See misalignment curve.		
Repeat Accuracy	10% of Sensing Range		
Output Current, Max.	200 mA (all outputs)		
Switching Current @ Voltage, Max.	24V DC +10%/-15%		
Operating Voltage/Supply Current	24V DC, +10%/-15% Class 2 SELV power supply		
Frequency of Operating Cycle	1 Hz		
Response Time (Off)	54 ms		
Outputs (Guard Door Closed, Actuator in Place)			
Safety Outputs	2 x PNP, 0.2 A, max.; Status: ON (+24V DC)		
Auxiliary Outputs	1 x PNP, 0.2 A max.; Status: OFF (0V DC)		
Environmental			
Operating Temperature—C (F)	-10...+55° (+14...+131°)		
Relative Humidity	5...95%		
Enclosure Type Rating	NEMA 3, 4X, 12, 13, IP 69K		
Shock	IEC68-2-27 30 g, 11 ms		
Vibration	IEC 68-2-6 10...55 Hz		
Radio Frequency	IEC 61000-4-3, IEC 61000-4-6		
Protection			
Protection Type	Short Circuit, Current Limitation, Overload, False Pulse, Transient Noise, Reverse Polarity, Overvoltage (inc. load dump), Thermal Shutdown with restart, Loss of GND and Vdd		
Loss of GND and Vdd protection	Incorporated		
Electrical Life	10 x 10 ⁶		

Safety Switches
Noncontact Switches
 SensaGuard™ 18 mm Barrel

Product Selection

Description	Operating Voltage/ Input Current	Safety Outputs	Auxiliary Outputs	Actuator Type	Assured Sensing Distance	Connection	Cat. No.
18 mm Plastic Barrel	24V DC, +10%/-15%/ 50 mA max.	2 Solid State outputs	1 Solid State output	18 mm	15 mm (0.59 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S16H
						3 m Cable	440N-Z21S16A
						10 m Cable	440N-Z21S16B
				30 mm	25 mm (0.98 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S26H
						3 m Cable	440N-Z21S26A
						10 m Cable	440N-Z21S26B
18 mm Stainless Steel Barrel	18 mm	10 mm (0.39 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S17H			
			3 m Cable	440N-Z21S17A			
			10 m Cable	440N-Z21S17B			

Recommended Logic Interfaces

Description	Safety Outputs	Auxiliary Outputs	Terminals	Reset Type	Power Supply	Cat. Pg. No.	Cat. No.
Single-Function Safety Relays							
MSR127RP	3 N.O.	1 N.C.	Removable Screw	Monitored Manual	24V AC/DC	—	440R-N23135
MSR127TP	3 N.O.	1 N.C.	Removable Screw	Automatic/Manual	24V AC/DC	—	440R-N23132
Modular Safety Relays							
MSR211P Base 2 N.C. only	2 N.O.	1 N.C. and 2 PNP Solid State	Removable	Automatic/Manual Monitored Manual	24V DC	—	440R-H23176
MSR220P Input Module	—	—	Removable	—	24V DC	—	440R-H23178
MSR310P Base	—	3 PNP Solid State	Removable	Automatic/Manual Monitored Manual	24V DC	—	440R-W23219
MSR320P Input Module	—	2 PNP Solid State	Removable	—	24V DC	—	440R-W23218

Note: For additional Safety Relays connectivity, see the Logic section of this catalog.
 For additional Safety I/O connectivity, see the Safety I/O section of this catalog.
 For Application and wiring diagrams, see the Applications section of this catalog.

Connection Systems

Description	Cat. No.
Cordset	889D-F8AB-*

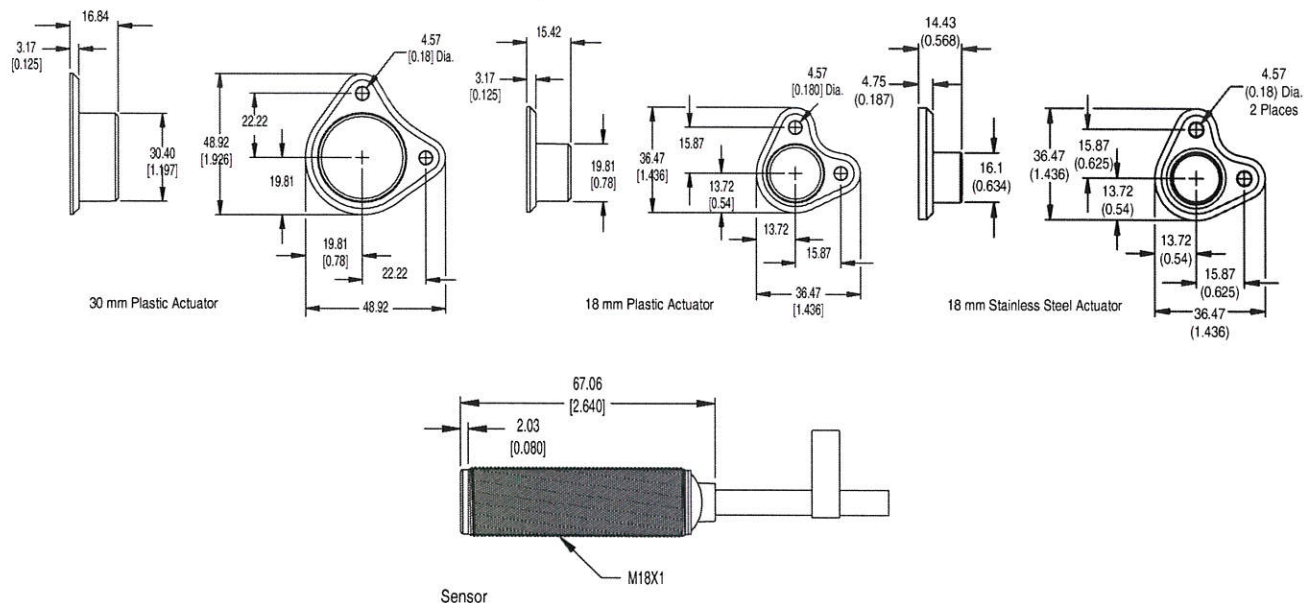
* Replace symbol with 2 (2 m), 5 (5 m), or 10 (10 m) for standard cable lengths.

Accessories

	Description	Dimensions	Cat. No.
	18 mm Plastic Actuator	—	440N-Z18PT
	30 mm Plastic Actuator	—	440N-Z30PT
	18 mm Stainless Steel Actuator	—	440N-Z18SST
	Mounting Bracket for Tubular Sensors—Right Angle Style	—	871A-BRS18
	Mounting Bracket for Tubular Sensors—Clamp Style	—	871A-BP18
	Snap Clamp Mounting Bracket	—	871A-SCBP18
	Swivel/Tilt bracket allows $\pm 10^\circ$ vertical and 360° rotation adjustment.	—	60-2649

Approximate Dimensions—mm (inches)

Dimensions are not intended to be used for installation purposes.



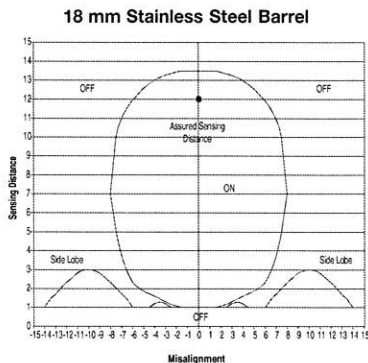
Noncontact Switches

Typical Wiring Diagrams

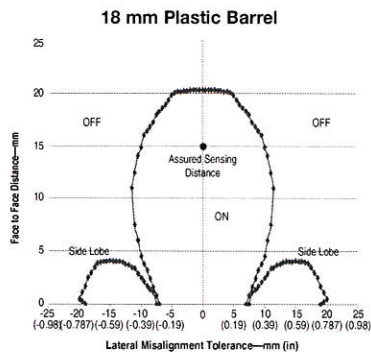
Description		Plastic		Stainless Steel	
8-Pin Micro (M12)					
8-Pin Cordset 889D-F8AB-*	Grey	OSSD 1	Safety A	OSSD 1	Safety A
	Red	OSSD 1+		OSSD 1+	
	Pink	OSSD 2	Safety B	OSSD 2	Safety B
	Yellow	OSSD 2+		OSSD 2+	
	White	Aux A		Aux A	
	Brown	24V DC	Unit Power	24V DC	Unit Power
	Blue	Gnd		Gnd	
Green	NA		Shield		

* Replace symbol with 2 (2 m), 5 (5 m) or 10 (10 m) for standard cable lengths.

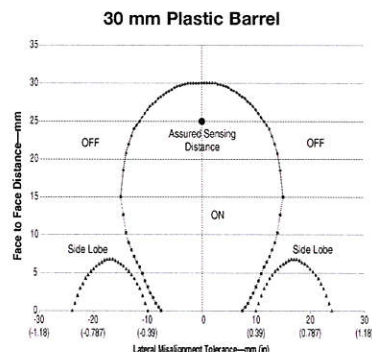
Misalignment Curve



Note: There must be a minimum spacing of 4 mm (0.157 in) if actuator and sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.

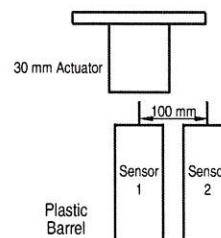
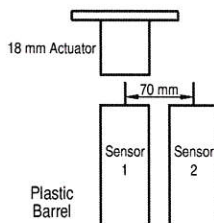
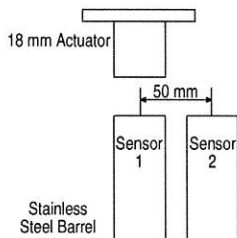


Note: There must be a minimum spacing of 4 mm (0.157 in) if actuator or sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.



Note: There must be a minimum spacing of 7 mm (0.275 in) if actuator and sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.

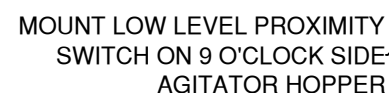
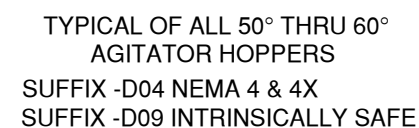
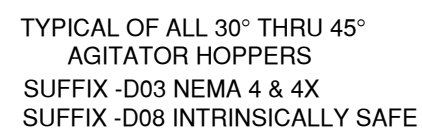
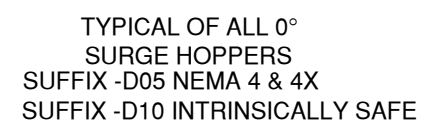
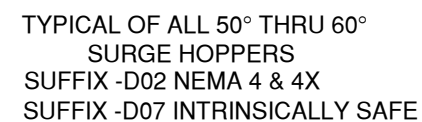
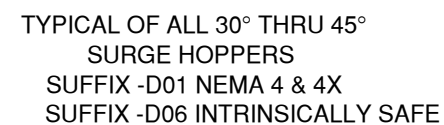
Minimum Distance Between Sensors



Diagnostic

Unit Indicators (per IEC 60073)

Device Output LED	State	Status	Troubleshooting
	Off	Not Powered	NA
	Red	Not Safe, OSSD Not Active	NA
	Green	Safe, OSSD Active	NA
	Green Flash	Power Up Test or OSSD Inputs Not Valid	Check 24V DC on OSSD Inputs (yellow and red wire)
	Red Flash	1 Hz Flash Recoverable Fault 4 Hz Flash Nonrecoverable Fault	Recoverable Fault: Check OSSD Outputs Are Not Shorted to GND, 24V DC or Each Other. Cycle Power.



REFERENCE 8220-
-D06, D07, D08, D09 & D10
FOR INTRINSICALLY SAFE

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE ENGLISH (FEET AND/OR INCHES)

TOLERANCES			
DECIMALS	INCHES	FEET	ANGLES
.X ± .030"	X" = ± 1/16"	X' - X" = ± 1/4" UP TO 20 FEET	± 1°
.XX ± .015"		X' - X" = ± 1/2" OVER 20 FEET	
.XXX ± .005"			

DO NOT SCALE DRAWING	REMOVE ALL SHARP EDGES
----------------------	------------------------

REV	REVISION DESCRIPTION	DATE	BY

HAPMAN
Ideas that move

P.O. BOX 2321
5944 EAST N AVE., KALAMAZOO, MICHIGAN 49048
PHONE (269) 343-1875 FAX (269) 349-2477

DATE	02/15/2021	JOB NO.	DRAWN C. WRIGHT	
SCALE	NONE		BOM = 0	DWG = 1

TITLE
LOW LEVEL PROXIMITY SWITCH, 24 VDC
MOUNTING ASSEMBLY

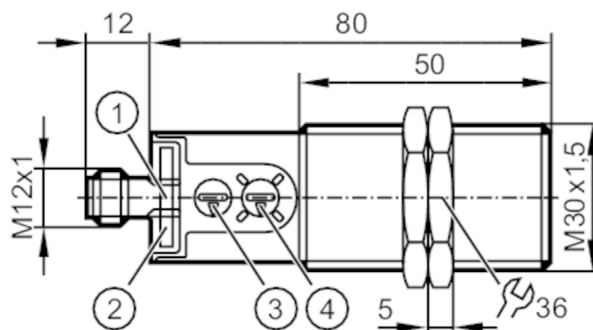
CUST.		
SHEET 1 OF 1	DWG. NO. 8239-D	REV.

8239-D



Capacitive sensor

KI-3250NFPKG/PL/2P/US/IO



- 1 LED yellow Switching status
- 2 LED green signal
- 3 potentiometer Sensing range
- 4 potentiometer Switching function



Product characteristics

Electrical design	PNP
Output function	normally open / closed; (selectable)
Sensing range [mm]	0.5...40
Communication interface	IO-Link
Housing	Threaded type
Dimensions [mm]	M30 x 1.5 / L = 92

Application

System	optical setting aid
Medium temperature [°C]	-25...110

Electrical data

Operating voltage [V]	10...30 DC
Current consumption [mA]	< 22
Protection class	III
Reverse polarity protection	yes

Outputs

Electrical design	PNP
Output function	normally open / closed; (selectable)
Max. voltage drop switching output DC [V]	2.5
Permanent current rating of switching output DC [mA]	200
Switching frequency DC [Hz]	30
Short-circuit protection	yes
Overload protection	yes



Capacitive sensor

KI-3250NFPKG/PL/2P/US/IO

Monitoring range		
Sensing range	[mm]	0.5...40
Sensing range adjustable		yes
Factory setting sensing range	[mm]	25
Real sensing range Sr	[mm]	25 ± 10 %
Accuracy / deviations		
Hysteresis	[% of Sr]	1...15
Switch-point drift	[% of Sr]	-10...10
Interfaces		
Communication interface		IO-Link
Transmission type		COM2 (38,4 kBaud)
IO-Link revision		1.1
SDCI standard		IEC 61131-9
Profiles		Smart Sensor: Process Data Variable; Device Identification
SIO mode		yes
Required master port class		A
Min. process cycle time	[ms]	20
IO-Link functions (acyclical)		application specific tag; operating hours counter; normally open / normally closed (configurable); PNP / NPN (parametrierbar); Switch point setting; Hysteresis
Operating conditions		
Ambient temperature	[°C]	-25...80
Protection		IP 65; IP 67; IP 69K
Tests / approvals		
EMC	EN 61000-4-2	30 kV AD / 4 kV CD
	EN 61000-4-3	10 V/m
	EN 61000-4-4	2 kV
	EN 61000-4-6	10 V
	EN 55011	class B
Vibration resistance	DIN EN 60947-5-2	1 mm amplitude, vibration duration 5 min., 30 min. per axis with resonance or 55 Hz
Shock resistance	DIN EN 60947-5-2	30 g 6 shocks / 11 ms half-sine (x, y, z)
MTTF	[years]	767.68
UL approval	UL approval number	D001
Mechanical data		
Weight	[g]	53.8
Housing		Threaded type
Mounting		non-flush mountable
Dimensions	[mm]	M30 x 1.5 / L = 92
Thread designation		M30 x 1.5
Material		PBT; PC; POM
Displays / operating elements		
Display	Switching status	1 x LED, yellow
	Power	1 x LED, green
	Setting aid	1 x LED, green
Optical setting aid		yes

KI6000



Capacitive sensor

KI-3250NFPKG/PL/2P/US/IO

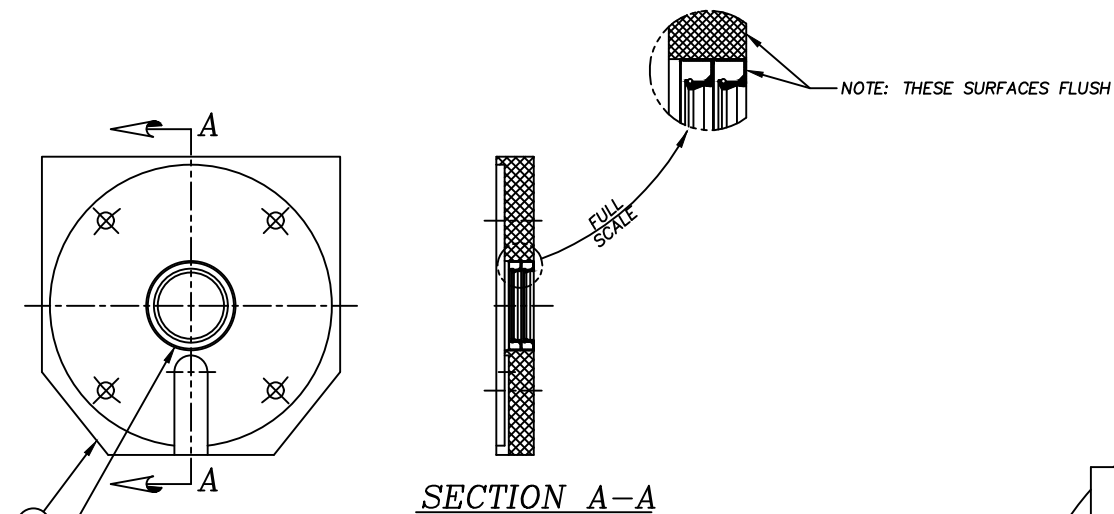
Accessories	
Items supplied	lock nuts: 2
Remarks	
Pack quantity	1 pcs.
Electrical connection - plug	

Connector: 1 x M12



Connection	
4:	OUT / IO-Link

REV	DESCRIPTION	DATE	BY
A	ITEM 3 WAS PART NO. X06318SD-800, DELETED ITEMS NO. 6 & 7.	10/23/14	MP
B	UPDATED TO DUAL SEAL, 3/8" S/B # SHOULDER BOLT	2/16/16	MP
C	ADDED 1/8" x 1" S/B FOR DRAINAGE	6/21/18	MP
D	REMOVED ITEM NO. 5 DETAIL, ALUM. SEAL PLATE, & SHEET 2	9/21/18	MP



SECTION A-A

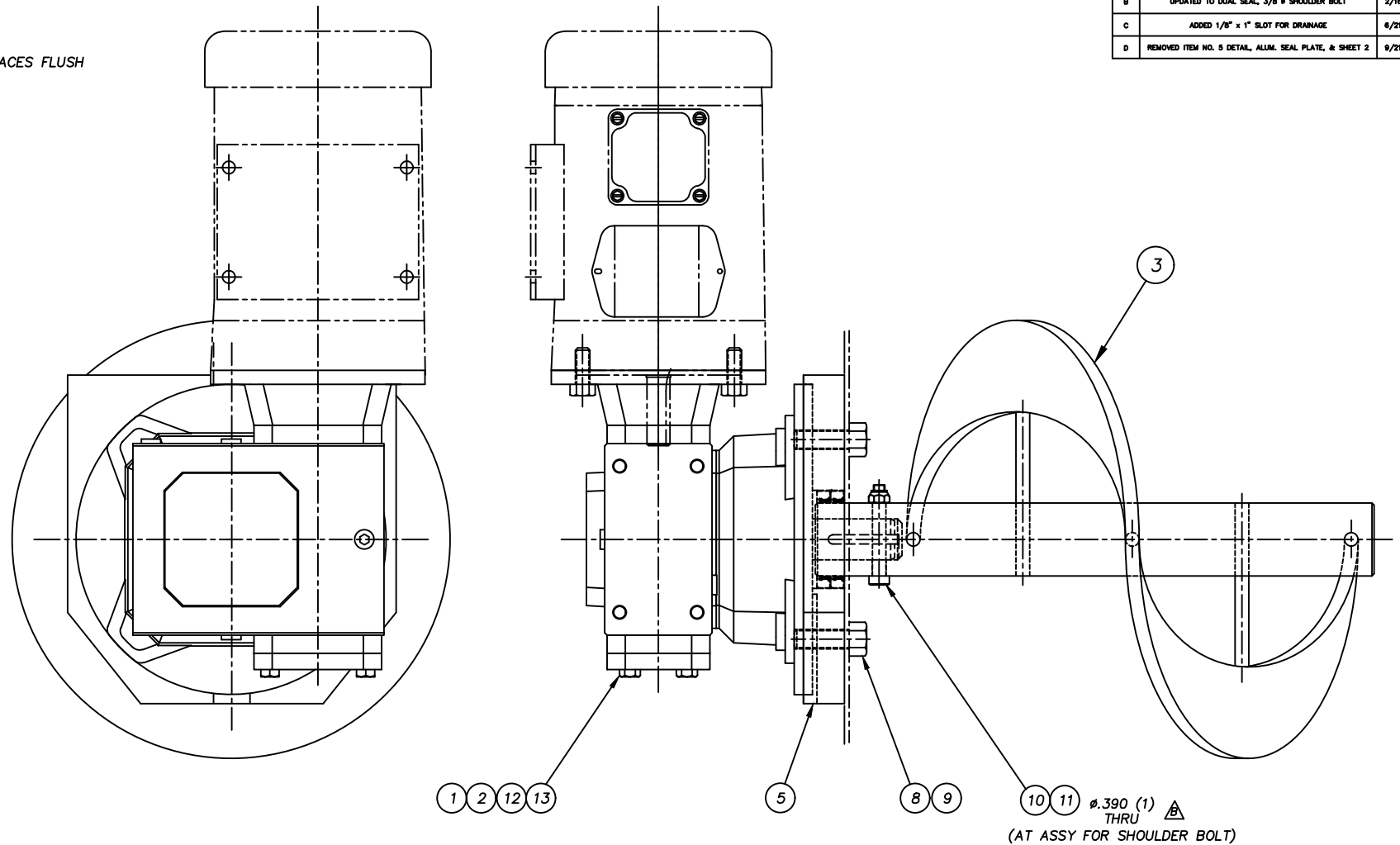
SEAL PLATE AND SEAL ASSY

SCALE: $3/8"=1"$

DWG #	ITEM #	HAPMAN ID/ PART #	DESCRIPTION	QTY
9563-D00 CARBON	1	GR-CMQ824-80-L-56C	REDUCER, GROVE	1
	2		MOTOR - REF - 56C, SEE HOPPER ASSY TOP LEVEL BOM	1
	3	X09762SD-D00	FLIGHTING, 12 DIA LH ELECTROPOLISHED 304SS X 12 LG X...	1
	4	19778	SEAL, LIP - CHICAGO RAWHIDE (OR EQUAL)	2
	5	X08437SD-D01	SEAL PLATE, 304/304L SS 1 1/8 THK MACHINED	1
	6			
	7			
	8		BOLT, HEX HD 1/2-13 X 1-1/4 LG, PLATED	4
	9		WASHER, LOCK 1/2 PLATED	4
	10	90298A632	SHOULDER BOLT, 3/8Ø SHOULDER, 2 LG, 18-8 SS	1
	11		NUT, NYLOCK #5/16-18 18-8 SS	1
	12		BOLT, HEX HD 3/8-16 X 1" LG, PLATED	4
	13		WASHER, FLAT 3/8 PLATED	4

DWG #	ITEM #	HAPMAN ID/ PART #	DESCRIPTION	QTY
9563-D01 304/304L SS	1	GR-CMQ824-80-L-56C	REDUCER, GROVE	1
	2		MOTOR - REF - 56C, SEE HOPPER ASSY TOP LEVEL BOM	1
	3	X09762SD-D01	FLIGHTING, 12 DIA LH ELECTROPOLISHED 304SS X 12 LG X...	1
	4	19778	SEAL, LIP - CHICAGO RAWHIDE (OR EQUAL)	2
	5	X08437SD-D01	SEAL PLATE, 304/304L SS 1 1/8 THK MACHINED	1
	6			
	7			
	8		BOLT, HEX HD 1/2-13 X 1-1/4 LG, SS	4
	9		WASHER, LOCK 1/2 SS	4
	10	90298A632	SHOULDER BOLT, 3/8Ø SHOULDER, 2 LG, 18-8 SS	1
	11		NUT, NYLOCK #5/16-18 18-8 SS	1
	12		BOLT, HEX HD 3/8-16 X 1" LG, PLATED	4

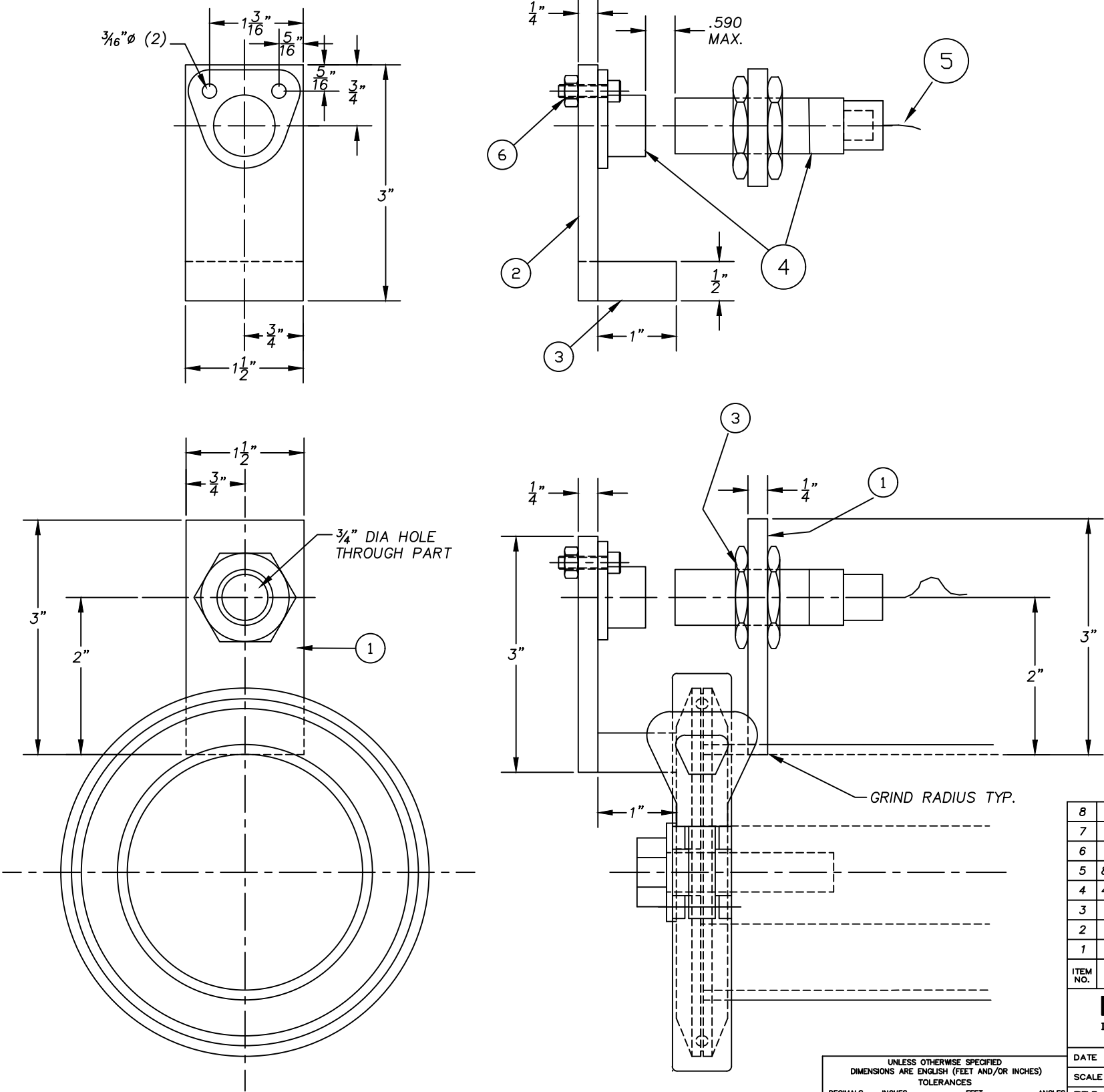
DWG #	ITEM #	HAPMAN ID/ PART #	DESCRIPTION	QTY
9563-D02 316/316L (AS NOTED)	1	GR-CMQ824-80-L-56C	REDUCER, GROVE	1
	2		MOTOR - REF - 56C, SEE HOPPER ASSY TOP LEVEL BOM	1
	3	X09762SD-D02	FLIGHTING, 12 DIA LH ELECTROPOLISHED 304SS X 12 LG X...	1
	4	19778	SEAL, LIP - CHICAGO RAWHIDE (OR EQUAL)	2
	5	X08437SD-D01	SEAL PLATE, 316/316L SS 1 1/8 THK MACHINED	1
	6			
	7			
	8		BOLT, HEX HD 1/2-13 X 1-1/4 LG, SS	4
	9		WASHER, LOCK 1/2 SS	4
	10	90298A632	SHOULDER BOLT, 3/8Ø SHOULDER, 2 LG, 18-8 SS	1
	11		NUT, NYLOCK #5/16-18 18-8 SS	1
	12		BOLT, HEX HD 3/8-16 X 1" LG, PLATED	4



AGITATOR/RIBBON DRIVE ASSY

		P.O. BOX 2321 8002 E. KILGORE ROAD, KALAMAZOO, MICHIGAN 49048 PHONE (269) 343-1875 FAX (269) 349-2477	
DATE	12/8/11	JOB NO.	
SCALE	3/8"=1"		
TITLE		DRAWN M. PORTER BOM = 0 DWG = 1	
AGITATOR/RIBBON ASSY, 12" 200-800 SERIES HELIX, 3/4 HP, 224, 80:1 STAINLESS STEEL BACKING PLATE			
CUST.			
SHEET	1 OF 1	DWG. NO.	9563-D _ _
			REV. <i>D</i>

REV	DESCRIPTION	DATE	BY
.	.	.	.



8			
7			
6	8-32 x 3/4 LG. SHCS W/LOCKNUT, SS		2
5	889D-F8AB-10 CABLE- 10 METER QC (AB)		1
4	440N-Z21S16H SAFETY SENSOR & ACTUATOR- 24V DC 18 mm (AB)		1
3	TARGET SPACER- 1/2 x 1 x 1 1/2, SS		1
2	TARGET PLATE- 1/4 x 1 1/2 x 3, SS		1
1	SWITCH PLATE- 1/4 x 1 1/2 x 3, SS		1
ITEM NO.	DRAWING NO.	DESCRIPTION	QUAN

HAPMAN Ideas that move		P.O. BOX 2321 6002 E. KILGORE ROAD, KALAMAZOO, MICHIGAN 49048 PHONE (269) 343-1675 FAX (269) 349-2477	
DATE	8/6/2020	JOB NO.	.
SCALE	FULL	DRAWN	M. PORTER
TITLE		BOM = 0	DWG = 1
FERRULE CAP SAFETY SENSOR SWITCH			
CUST.		SHEET	1 OF 1
DWG. NO.	8218-C	REV.	

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE ENGLISH (FEET AND/OR INCHES) TOLERANCES			
DECIMALS	INCHES	FEET	ANGLES
.X ±.030		X'-X"=±1/4" UP TO 20 FEET	
.XX ±.015		X'-X"=±1/2" OVER 20 FEET	±1°
.XXX ±.005	X"= ±1/16"		
DO NOT SCALE DWG. BREAK ALL SHARP EDGES			
MATERIAL			
FINISH			



Description

When it comes to machine safety, Rockwell Automation knows that protection of personnel and equipment is your main concern. At the same time, flexibility and productivity are points that must also be considered as you design your safety system. Optimize all of these with the new Allen-Bradley SensaGuard family of noncontact switches.

Featuring the latest generation of RFID technology for coding and inductive technology for sensing, SensaGuard's large sensing range and tolerance to misalignment is a cost-effective solution that is ideally suited for a wide range of industrial safety applications.

The SensaGuard product line is a Category 4 /SIL 3 rated switch per EN954-1, TÜV functional safety approved to IEC 61508.

Features

- Switches can be connect to a standard safety relay, for example, the MSR126, MSR127, MSR200/300 Family, SmartGuard and Safety I/O Blocks
- Multiple actuator sizes for large sensing distance
- IP 69K environmental rating
- Short-circuit and over-voltage protection
- Led located on the switch for door status and troubleshooting

Benefits

- No dedicated controller required
- Cat 4/SIL 3 rating maintained even with multiple units connected in series
- Switches can be connected in series with other devices (light curtain, E-stops, key interlock switches)
- Extended diagnostics for easy troubleshooting
- Large sensing distances
- Tolerance to misalignment
- Multiple sensing directions
- Stainless steel version suitable for use in harsh environments
- Use standard proximity brackets

Specifications

Category	Cat. 4/SIL3		
Certifications	TÜV, CE, cULus (UL 508)		
Standards	IEC 60947-5-3, IEC 61508, EN 954		
PFHd	1.119...10 ⁻⁹		
Operating Characteristics			
Sensing Distance (Target)	18 mm Plastic	30 mm Plastic	18 mm SS
Operating Distance, Make—mm (in)	Assured: 15 (0.59)	Assured: 25 (0.98)	Assured: 10 (0.39)
Case Material	Valox® DR 48		304 Stainless Steel
Actuator Material	Valox® DR 48		304 Stainless Steel
Typical Misalignment	See misalignment curve.		
Repeat Accuracy	10% of Sensing Range		
Output Current, Max.	200 mA (all outputs)		
Switching Current @ Voltage, Max.	24V DC +10%/-15%		
Operating Voltage/Supply Current	24V DC, +10%/-15% Class 2 SELV power supply		
Frequency of Operating Cycle	1 Hz		
Response Time (Off)	54 ms		
Outputs (Guard Door Closed, Actuator in Place)			
Safety Outputs	2 x PNP, 0.2 A, max.; Status: ON (+24V DC)		
Auxiliary Outputs	1 x PNP, 0.2 A max.; Status: OFF (0V DC)		
Environmental			
Operating Temperature—C (F)	-10...+55° (+14...+131°)		
Relative Humidity	5...95%		
Enclosure Type Rating	NEMA 3, 4X, 12, 13, IP 69K		
Shock	IEC68-2-27 30 g, 11 ms		
Vibration	IEC 68-2-6 10...55 Hz		
Radio Frequency	IEC 61000-4-3, IEC 61000-4-6		
Protection			
Protection Type	Short Circuit, Current Limitation, Overload, False Pulse, Transient Noise, Reverse Polarity, Overvoltage (inc. load dump), Thermal Shutdown with restart, Loss of GND and Vdd		
Loss of GND and Vdd protection	Incorporated		
Electrical Life	10 x 10 ⁶		

Safety Switches
Noncontact Switches
 SensaGuard™ 18 mm Barrel

Product Selection

Description	Operating Voltage/ Input Current	Safety Outputs	Auxiliary Outputs	Actuator Type	Assured Sensing Distance	Connection	Cat. No.
18 mm Plastic Barrel	24V DC, +10%/-15%/ 50 mA max.	2 Solid State outputs	1 Solid State output	18 mm	15 mm (0.59 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S16H
						3 m Cable	440N-Z21S16A
						10 m Cable	440N-Z21S16B
				30 mm	25 mm (0.98 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S26H
						3 m Cable	440N-Z21S26A
						10 m Cable	440N-Z21S26B
18 mm Stainless Steel Barrel	18 mm	10 mm (0.39 in)	6 in Pigtail, 8-Pin Micro (M12)	440N-Z21S17H			
			3 m Cable	440N-Z21S17A			
			10 m Cable	440N-Z21S17B			

Recommended Logic Interfaces

Description	Safety Outputs	Auxiliary Outputs	Terminals	Reset Type	Power Supply	Cat. Pg. No.	Cat. No.
Single-Function Safety Relays							
MSR127RP	3 N.O.	1 N.C.	Removable Screw	Monitored Manual	24V AC/DC	—	440R-N23135
MSR127TP	3 N.O.	1 N.C.	Removable Screw	Automatic/Manual	24V AC/DC	—	440R-N23132
Modular Safety Relays							
MSR211P Base 2 N.C. only	2 N.O.	1 N.C. and 2 PNP Solid State	Removable	Automatic/Manual Monitored Manual	24V DC	—	440R-H23176
MSR220P Input Module	—	—	Removable	—	24V DC	—	440R-H23178
MSR310P Base	—	3 PNP Solid State	Removable	Automatic/Manual Monitored Manual	24V DC	—	440R-W23219
MSR320P Input Module	—	2 PNP Solid State	Removable	—	24V DC	—	440R-W23218

Note: For additional Safety Relays connectivity, see the Logic section of this catalog.
 For additional Safety I/O connectivity, see the Safety I/O section of this catalog.
 For Application and wiring diagrams, see the Applications section of this catalog.

Connection Systems

Description	Cat. No.
Cordset	889D-F8AB-*

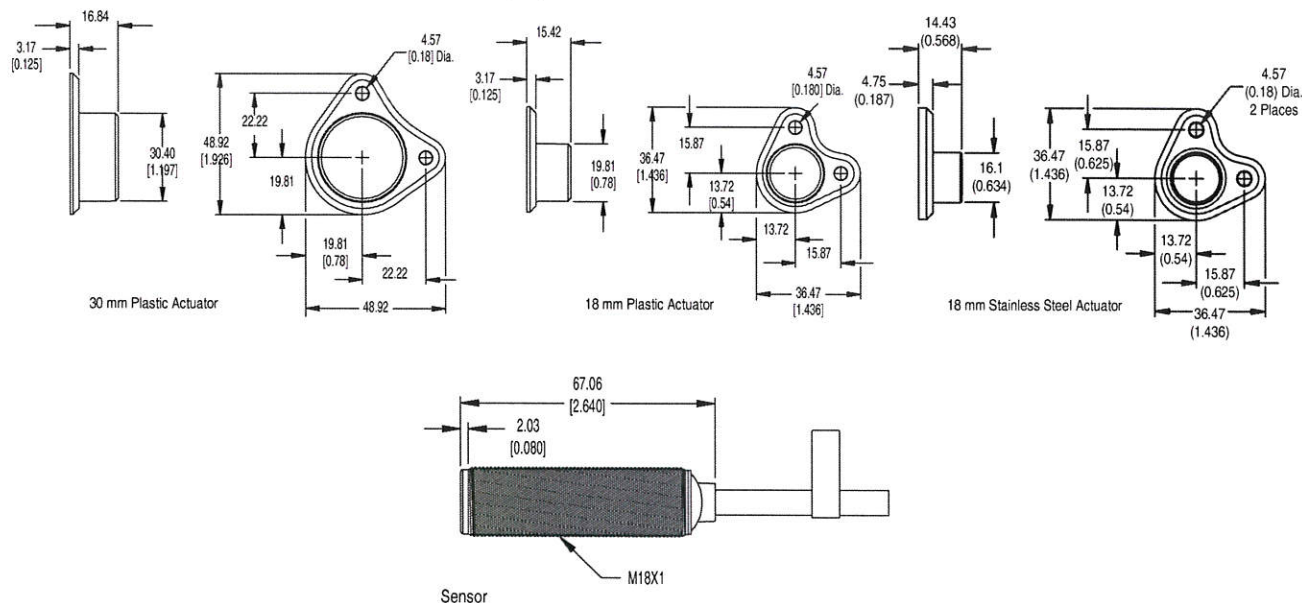
* Replace symbol with 2 (2 m), 5 (5 m), or 10 (10 m) for standard cable lengths.

Accessories

	Description	Dimensions	Cat. No.
	18 mm Plastic Actuator	—	440N-Z18PT
	30 mm Plastic Actuator	—	440N-Z30PT
	18 mm Stainless Steel Actuator	—	440N-Z18SST
	Mounting Bracket for Tubular Sensors—Right Angle Style	—	871A-BRS18
	Mounting Bracket for Tubular Sensors—Clamp Style	—	871A-BP18
	Snap Clamp Mounting Bracket	—	871A-SCBP18
	Swivel/Tilt bracket allows $\pm 10^\circ$ vertical and 360° rotation adjustment.	—	60-2649

Approximate Dimensions—mm (inches)

Dimensions are not intended to be used for installation purposes.

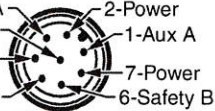
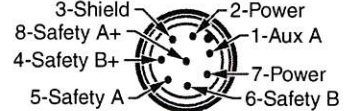


Safety Switches

Noncontact Switches

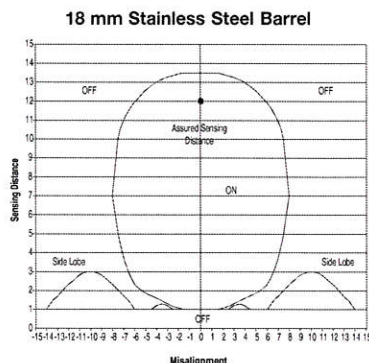
SensaGuard™ 18 mm Barrel

Typical Wiring Diagrams

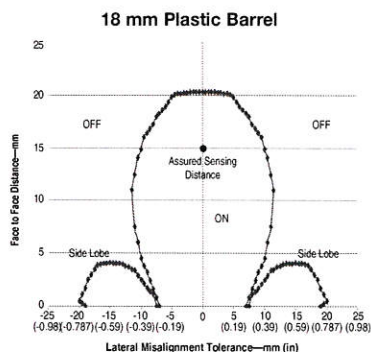
Description		Plastic		Stainless Steel	
8-Pin Micro (M12)					
8-Pin Cordset 889D-F8AB-*	Grey	OSSD 1	Safety A	OSSD 1	Safety A
	Red	OSSD 1+		OSSD 1+	
	Pink	OSSD 2	Safety B	OSSD 2	Safety B
	Yellow	OSSD 2+		OSSD 2+	
	White	Aux A		Aux A	
	Brown	24V DC	Unit Power	24V DC	Unit Power
	Blue	Gnd		Gnd	
	Green	NA		Shield	

* Replace symbol with 2 (2 m), 5 (5 m) or 10 (10 m) for standard cable lengths.

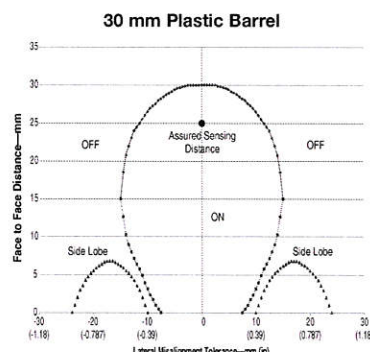
Misalignment Curve



Note: There must be a minimum spacing of 4 mm (0.157 in) if actuator and sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.

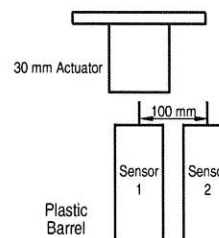
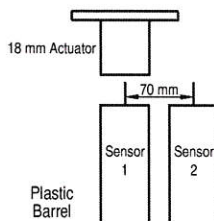
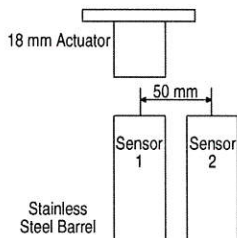


Note: There must be a minimum spacing of 4 mm (0.157 in) if actuator and sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.



Note: There must be a minimum spacing of 7 mm (0.275 in) if actuator and sensor face approaches laterally. This will prevent false triggering due to the side lobe areas.

Minimum Distance Between Sensors



Diagnostic

Unit Indicators (per IEC 60073)

Device Output LED	State	Status	Troubleshooting
	Off	Not Powered	NA
	Red	Not Safe, OSSD Not Active	NA
	Green	Safe, OSSD Active	NA
	Green Flash	Power Up Test or OSSD Inputs Not Valid	Check 24V DC on OSSD Inputs (yellow and red wire)
	Red Flash	1 Hz Flash Recoverable Fault 4 Hz Flash Nonrecoverable Fault	Recoverable Fault: Check OSSD Outputs Are Not Shorted to GND, 24V DC or Each Other. Cycle Power.